Reactive vs Preemptive Vaccination Strategies

AUTHOR
Jong-Hoon Kim

PUBLISHED
July 5, 2025

1 Setup

2 Introduction

In the face of emerging infectious disease threats, public health officials must decide whether to deploy vaccines preemptively (before an outbreak occurs) or reactively (after an outbreak is detected). This strategic choice carries high stakes for both health outcomes and resource utilization. Preemptive vaccination can avert outbreaks by immunizing at-risk populations in advance, but it incurs upfront costs and risks vaccinating for an outbreak that might never occur. Reactive vaccination, in contrast, targets outbreaks once they are detected, potentially saving doses and avoiding unnecessary campaigns; however, losses accrue before control measures take effect (Middleton and Larremore 2024; Azman et al. 2019).

This dilemma is complicated by uncertainty. Outbreak risks vary between populations due to heterogeneous social, environmental, and biological factors, yet risk forecasts are imperfect (Wells et al. 2020). If authorities had perfect information, they could vaccinate high-risk populations in advance. In practice, risk estimates may be noisy, leading to misallocation of limited vaccines and potential failure to prevent major outbreaks (Colijn, Worby, and Lipsitch 2021).

Models from diverse diseases highlight that no one strategy is best across settings. Studies on mpox, hepatitis E, and MERS-CoV suggest that preemptive campaigns can be life-saving if outbreaks are severe or grow quickly (Quilty et al. 2023; Azman et al. 2019; Wells et al. 2020). Conversely, reactive strategies may use doses more efficiently when supply is constrained or preemptive coverage is uncertain (Middleton and Larremore 2024).

Despite these insights, no unifying quantitative framework exists to compare reactive and preemptive vaccination under varying outbreak probability, resource availability, and risk uncertainty. Our study addresses this gap by developing an analytic and simulation-based framework for identifying the optimal allocation of vaccine resources — defined as the proportion α used preemptively — across populations with potentially heterogeneous and uncertain risk.

Through analytical derivation and simulation, we characterize the impact of outbreak probability (p), reactive efficacy (r), vaccine budget (f), and targeting accuracy (Spearman rank correlation $ ho$) on strategy performance. Our results offer practical guidance for policymakers deciding how much to invest in preemptive campaigns when vaccine stockpiles are constrained and risk signals are imperfect.

3 The Model

3.1 The Single Population Scenario

We evaluate the expected **total cost** of two strategies (preemptive vs reactive vaccination) for **a single population** at risk of an outbreak.

3.1.1 Parameters

Let:

- D: The full cost of an outbreak (e.g., cases, deaths, DALYs)
- C: The cost of deploying a vaccination campaign
- p: The probability of an outbreak ($0 < p < 1$)
- r: The **relative effectiveness** of a reactive vaccination campaign compared to the preemptive vaccination campaign ($0 < r < 1$)

3.2 Strategy 1: Preemptive Vaccination

- We vaccinate regardless of whether an outbreak happens
- Outbreak is fully averted if it would have occurred
- Always pay vaccination cost C
- Expected outbreak cost is 0

$$\text{Cost}_{\text{pre}} = C$$

3.3 Strategy 2: Reactive Vaccination

- Vaccinate only if an outbreak occurs (probability p)
- Outbreak is only partially mitigated: remaining cost is $(1-r)D$
- Vaccination cost C is also incurred only when outbreak happens

$$\text{Cost}_{\text{react}} = p \cdot (C + (1-r)D)$$

3.4 Comparing the two strategies

We prefer **preemptive vaccination** when:

$$C < p \cdot (C + (1-r)D)$$

Solving for p:

$$p^* = \frac{C}{C + (1-r)D}$$

This is the **threshold outbreak probability** above which preemptive vaccination is cost-saving.

3.4.1 Interpretation

- As $r \to 1$, reactive vaccination becomes nearly as effective as preemptive -> threshold p^* increases
- As the cost of an outbreak compared to the vaccination cost, D/C, becomes large -> threshold p^* decreases

3.5 Symbolic derivation

▶ Code

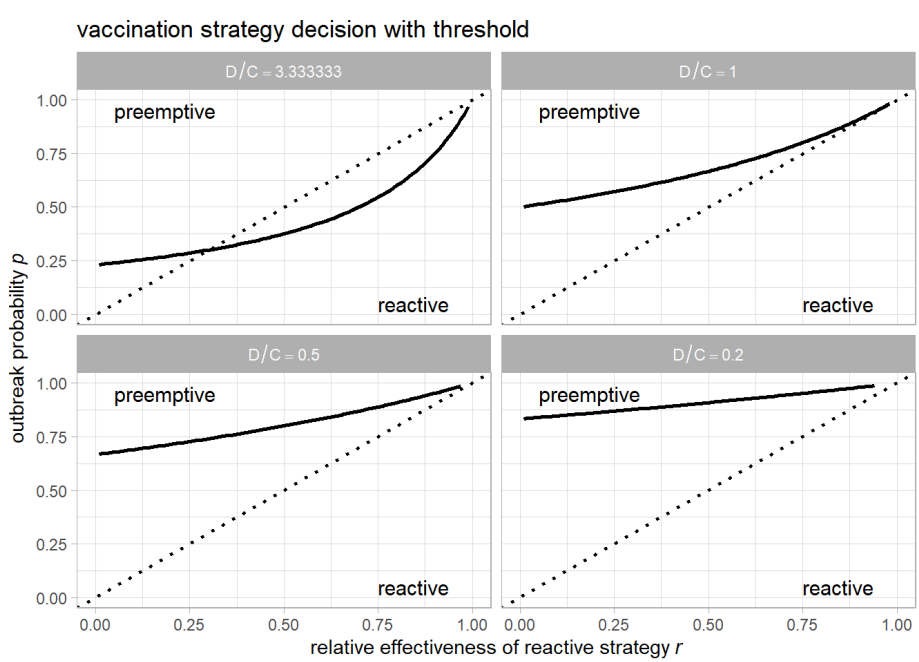
```
C/(C - D*r + D)
```

Expected output:

$$\frac{C}{C + (1-r)D}$$

3.6 Visualizing the threshold p^*

▶ Code



3.7 Summary

This analysis shows that **preemptive vaccination** is preferred when:

$$p > \frac{C}{C + (1-r)D}$$

This tradeoff depends jointly on: - **Probability of an outbreak** (p), - **Relative effectiveness of a reactive strategy** (r), - **Cost of prevention** (C), - **Cost of inaction** (D)

4 Two-population scenario

We assume:

- Two populations, each has an independent probability p of outbreak
- Only one vaccination campaign is available (can be used **preemptively** or **reactively** for one population)
- Key parameters:

Symbol	Description
X	Number of outbreaks ($X = 0, 1, 2$)
p	Probability of outbreak in a population
C	Cost of one vaccination campaign
D	Cost of one unmitigated outbreak
r	Relative effectiveness of **reactive** vaccination ($0 < r < 1$)

4.1 Preemptive strategy

We vaccinate **one population in advance**, say population A. No reactive response is used.

4.1.1 Case 0: No outbreaks ($X = 0$)

- Probability: $(1-p)^2$
- Cost: only unnecessary vaccination of A → C

$$\text{Cost}_{X=0,\text{pre}} = C$$

Case 1: One outbreak ($X = 1$)

- Probability: $2p(1-p)$
- Two subcases (each with prob. $p(1-p)$):
  - If outbreak occurs in A (vaccinated): outbreak prevented -> cost = C
  - If outbreak occurs in B (unvaccinated): outbreak occurs -> cost = $C + D$

Average cost:

$$\text{Cost}_{X=1,\text{pre}} = 2p(1-p)\left(C + \frac{D}{2}\right)$$

Case 2: Two outbreaks ($X = 2$)

- Probability: p^2
- One outbreak prevented (in vaccinated population), one unmitigated -> cost = $C + D$

$$\text{Cost}_{X=2,\text{pre}} = p^2(C + D)$$

4.1.2 Total expected cost for the preemptive strategy

$$\begin{aligned}
\mathbb{E}[\text{Cost}_{\text{pre}}] &= (1-p)^2 C + 2p(1-p)\left(C + \frac{D}{2}\right) + p^2(C + D) \\
&= C(1-p)^2 + 2p(1-p)C + p(1-p)D + p^2 C + p^2 D \\
&= C[(1-p)^2 + 2p(1-p) + p^2] + D[p(1-p) + p^2] \\
&= C + pD
\end{aligned}$$

4.2 Reactive strategy

We wait for outbreaks and respond to **one** if detected.

4.2.1 Case 0: No outbreaks ($X = 0$)

- Probability: $(1-p)^2$
- No vaccination needed, no outbreak → cost = 0

$$\text{Cost}_{X=0,\text{react}} = 0$$

4.2.2 Case 1: One outbreak ($X = 1$)

- Probability: $2p(1-p)$
- Detect and respond to outbreak:
  - Vaccination cost = C
  - Partial mitigation: residual cost = $(1-r)D$

$$\text{Cost}_{X=1,\text{react}} = 2p(1-p)\left(C + (1-r)D\right)$$

4.2.3 Case 2: Two outbreaks ($X = 2$)

- Probability: p^2
- Only one outbreak can be addressed:
  - Vaccination cost = C
  - One outbreak mitigated -> $(1-r)D$
  - One outbreak unmitigated -> D

$$\text{Cost}_{X=2,\text{react}} = p^2\left(C + (1-r)D + D\right) = p^2(C + (2-r)D)$$

4.2.4 Total Expected Cost – Reactive Strategy

$$\begin{aligned}\mathbb{E}[\text{Cost}_{\text{react}}] &= (1-p)^2 \cdot 0 + 2p(1-p)\left(C + (1-r)D\right) + p^2\left(C + (2-r)D\right) \\ &= 2p(1-p)\left(C + (1-r)D\right) + p^2\left(C + (2-r)D\right)\end{aligned}$$

4.3 Symbolic Comparison

Compare:

$$\mathbb{E}[\text{Cost}_{\text{pre}}] = C + pD$$

vs.

$$\mathbb{E}[\text{Cost}_{\text{react}}] = (2p - p^2)C + \left[2p(1-p)(1-r) + p^2(2-r)\right]D$$

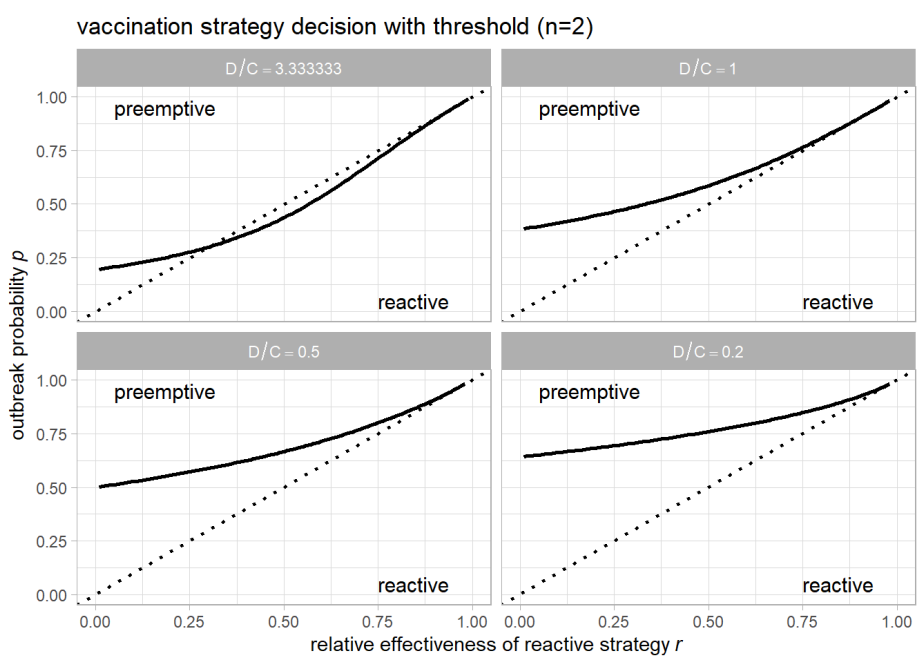
Preemptive strategy is preferred when:

$$C + pD < (2p - p^2)C + \left[2p(1-p)(1-r) + p^2(2-r)\right]D$$

▶ Code

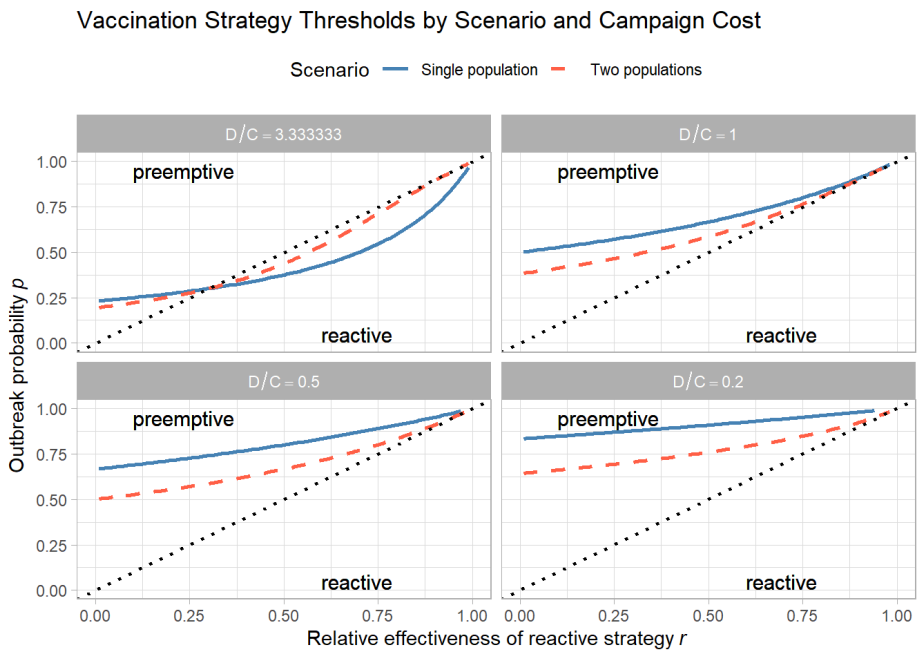
```
Threshold p solution 1:
(C - sqrt(D)*sqrt(-4*C*r + 4*C + 4*D*r**2 - 4*D*r + D)/2 - D*r + D/2)/(C - D*r)
Threshold p solution 2:
(C + sqrt(D)*sqrt(-4*C*r + 4*C + 4*D*r**2 - 4*D*r + D)/2 - D*r + D/2)/(C - D*r)
```

▶ Code



4.4 Compare single and two population scenarios

▶ Code



4.5 Multiple populations ($n > 2$) scenario

We consider n populations where each has probability p of an outbreak. We can afford to vaccinate a fraction f of them, pre-emptively or reactively.

Let:

- n: number of populations
- p: outbreak probability
- f: fraction of populations we can vaccinate
- C: vaccination cost per population
- D: outbreak cost (if unmitigated)
- r: relative effectiveness of reactive vaccination ($0 < r < 1$)

4.5.1 Strategy 1: Preemptive Vaccination

We preemptively vaccinate fn populations. If an outbreak occurs in the $(1-f)n$ unvaccinated group, we pay D.

Total expected cost:

$$\text{Cost}_{\text{pre}} = fn \cdot C + (1-f)np \cdot D$$

4.5.2 Reactive vaccination strategy

4.6 Step 2: Conditioning on $X = x$ outbreaks

For a fixed number of outbreaks $X = x$:

- If $x \leq k$, we can respond to **all** outbreaks.

  Total cost:

  $$\text{Cost}_{x,\text{react}} = x(C + (1-r)D)$$

- If $x > k$, we can respond to k outbreaks, and $x - k$ go **unmitigated**.

  Mitigated cost:

  $$k(C + (1-r)D)$$

  Unmitigated cost:

  $$(x-k)D$$

So the **expected cost when** $X = x$ is:

$$\text{Cost}_{x,\text{react}} = \begin{cases} x(C + (1-r)D), & \text{if } x \leq k \\ k(C + (1-r)D) + (x-k)D, & \text{if } x > k \end{cases}$$

4.7 Step 3: Total Expected Cost

We now compute the total expected cost across all possible x:

$$\mathbb{E}[\text{Cost}_{\text{react}}] = \sum_{x=0}^{n} \Pr(X = x) \cdot \text{Cost}_{x,\text{react}}$$

Split into two parts:

4.7.1 Case 1: $x \leq k$

$$\sum_{x=0}^{k} \binom{n}{x} p^x (1-p)^{n-x} \cdot x(C + (1-r)D)$$

4.7.2 Case 2: $x > k$

$$\sum_{x=k+1}^{n} \binom{n}{x} p^x (1-p)^{n-x} \cdot [k(C + (1-r)D) + (x-k)D]$$

This expression does **not simplify into closed form**, but it can be computed numerically or approximated.

4.7.3 Strategy 2-1: Reactive Vaccination (Approximate)

We reactively vaccinate up to fn populations **after** outbreaks occur.

Expected number of outbreaks: np
We can respond to $\min(np, fn)$ and leave $\max(np - fn, 0)$ unmitigated.

Residual cost per mitigated outbreak = $(1-r)D$

Expected cost:

$$\text{Cost}_{\text{react}} = \min(np, fn)(C + (1-r)D) + \max(np - fn, 0) \cdot D$$

4.8 Symbolic Comparison using SymPy

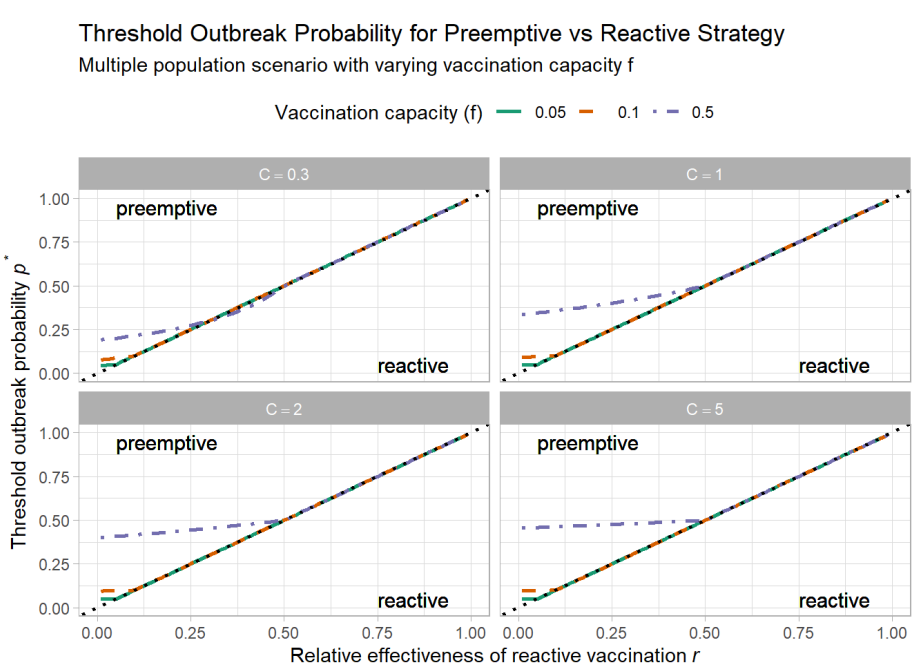
▶ Code

```
-C*f*n + D*n*p*(f - 1) + D*Max(0, n*(-f + p)) + (C - D*(r - 1))*Min(f*n, n*p)
```

4.9 Decision Thresholds for Preemptive vs Reactive (Multiple Populations)

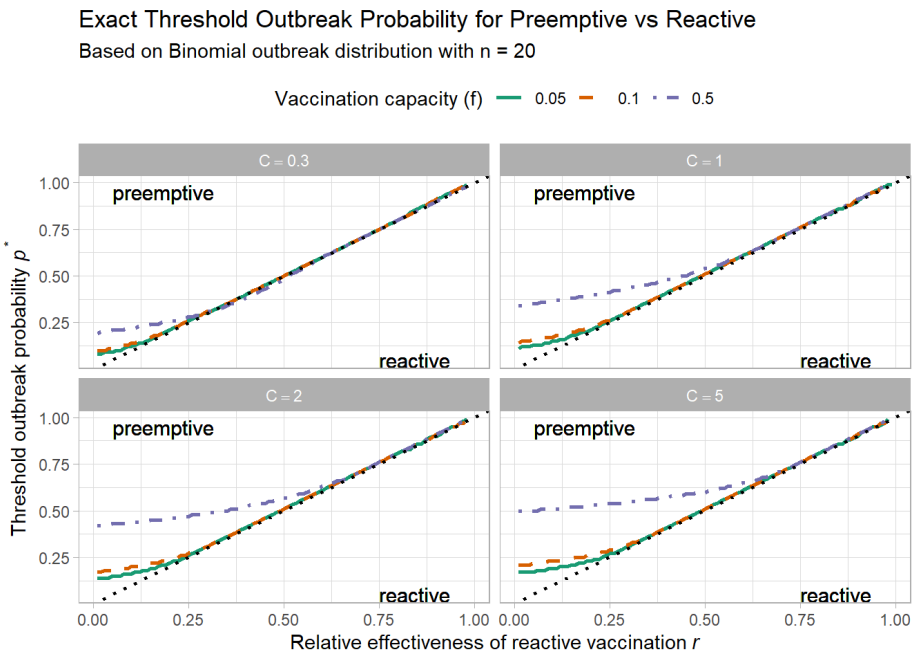
We compute and plot threshold outbreak probability p^* at which preemptive vaccination becomes cheaper than reactive, under various vaccination cost levels C and vaccination capacity levels f.

▶ Code

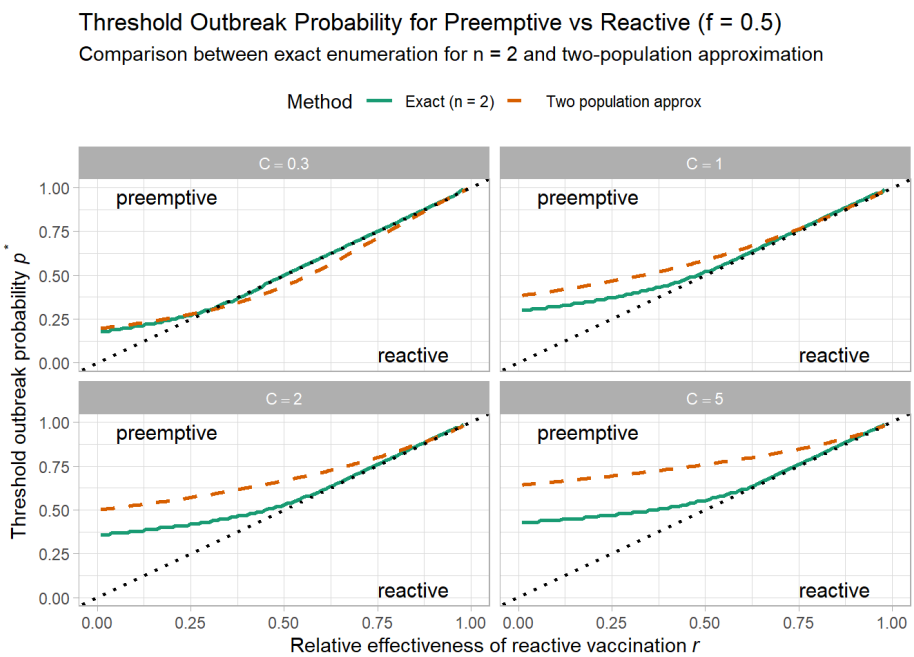


▶ Code

▶ Code



▶ Code



4.10 Asymptotic Limit: Preemptive vs Reactive

As the number of populations $n \to \infty$ and vaccination capacity fraction $f \to 0$ (i.e., resources are extremely limited), the expected costs for preemptive and reactive strategies simplify.

Let:

- p: probability of an outbreak per population,
- r: relative effectiveness of reactive vaccination,
- C: cost of one vaccination campaign,
- D: cost of one unmitigated outbreak,
- fn: number of populations we can afford to vaccinate (preemptively or reactively)

4.10.1 Preemptive Expected Cost

Total cost:

$$\text{Cost}_{\text{pre}} = fn \cdot C + (1 - f)np \cdot D$$

As $f \to 0$, this becomes:

$$\text{Cost}_{\text{pre}} \approx npD$$

4.10.2 Reactive Expected Cost

Using the expression:

$$\text{Cost}_{\text{react}} = \min(fn, np)(C + (1-r)D) + \max(np - fn, 0) \cdot D$$

If $fn \ll np$:

$$\text{Cost}_{\text{react}} \approx fn(C + (1-r)D) + (np - fn)D$$

4.10.3 Cost Difference

Compute:

$$\begin{aligned}\Delta &= \text{Cost}_{\text{pre}} - \text{Cost}_{\text{react}} \\ &= fnC + (1-f)npD - [fn(C + (1-r)D) + (np - fn)D] \\ &= fnD(r - p)\end{aligned}$$

4.10.4 Result

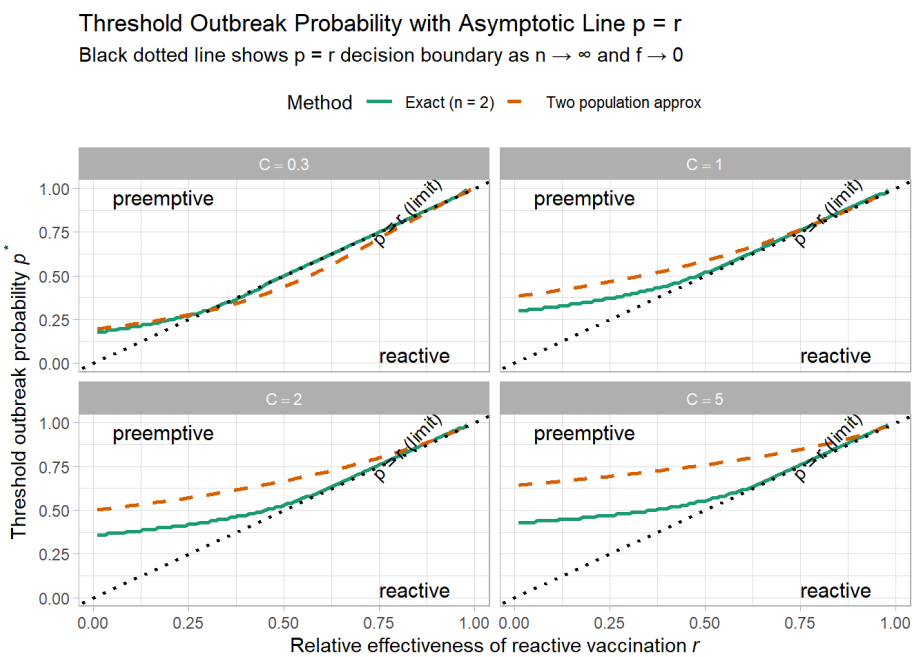
- If $r < p$: preemptive is cheaper ($\Delta < 0$)
- If $r > p$: reactive is cheaper ($\Delta > 0$)

Thus, in the limit:

$$p > r \quad \Rightarrow \quad \text{preemptive is preferred}$$

This corresponds to the decision threshold line $p^* = r$ as $f \to 0$ and $n \to \infty$.

▶ Code



5 Optimizing Hybrid Vaccination Strategy

5.1 Problem

> Given a limited vaccine supply, how should we split our efforts between **preemptive** and **reactive** vaccination strategies to **minimize total expected cost**?

We define:

- $\alpha \in [0, 1]$: fraction of vaccination resources used **preemptively**
- $1 - \alpha$: fraction used **reactively**
- f: total fraction of the population that can be vaccinated
- n: total number of populations
- p: probability of outbreak in each population

- C: cost of one vaccination campaign
- D: cost of one unmitigated outbreak
- r: relative effectiveness of reactive vaccination to the preemptive strategy

5.2 Expected Cost Formula

The total expected cost of a hybrid strategy that uses α of the vaccination capacity preemptively is:

$$\text{Cost}(\alpha) = \underbrace{\alpha f n C}_{\text{preemptive vaccination}} + \underbrace{\min(x, k_{\text{react}})(C + (1-r)D)}_{\text{reactive cost}} + \underbrace{\max(x - k_{\text{react}}, 0)D}_{\text{unmitigated outbreaks}}$$

Where: - $x = (1 - \alpha f)np$: expected number of outbreaks in unvaccinated group - $k_{\text{react}} = (1-\alpha)fn$: number of reactive campaigns available

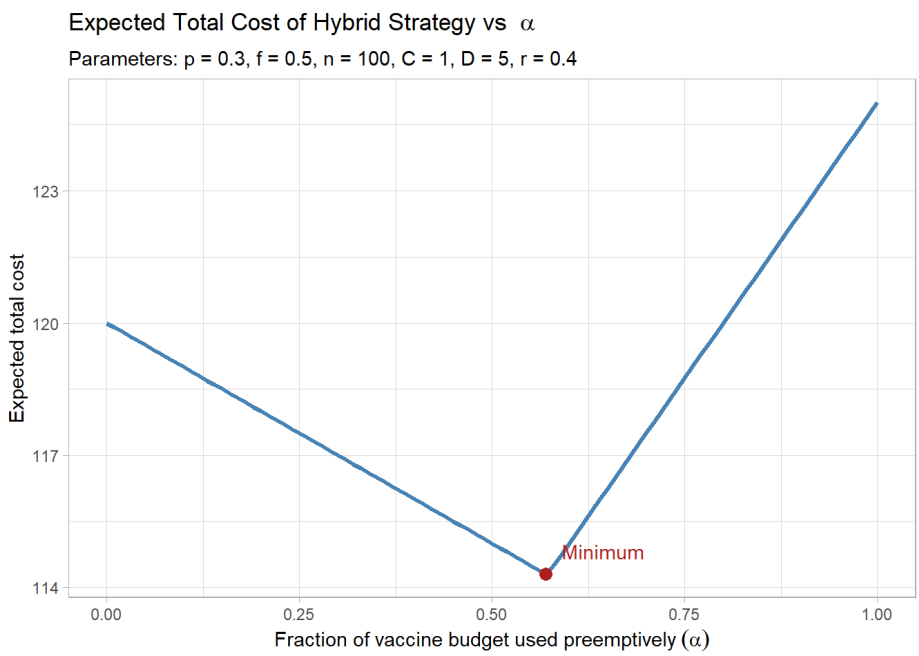
5.2.1 Visualizing Cost of Hybrid Strategy

We explore how the total expected cost varies with the fraction α of vaccine budget used preemptively.

▶ Code

Plot

▶ Code

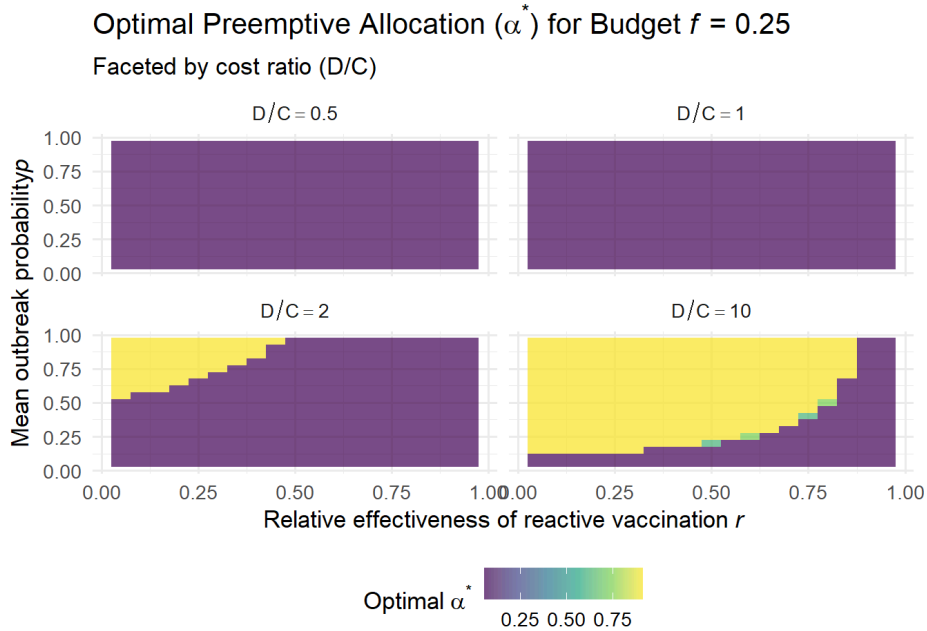


Optimum α

▶ Code

Plot

▶ Code



5.3 Heterogeneous risk

5.3.1 Beta distribution

Alternative parameterization of $\mathrm{Beta}(a, b)$ is using mean p and the sample size (or concentration)

$$a = p\kappa$$

,

$$b = (1-p)\kappa$$

$$\mathrm{Var}(X) = \frac{p(1-p)}{1+\kappa}$$

Find optimal α

▶ Code

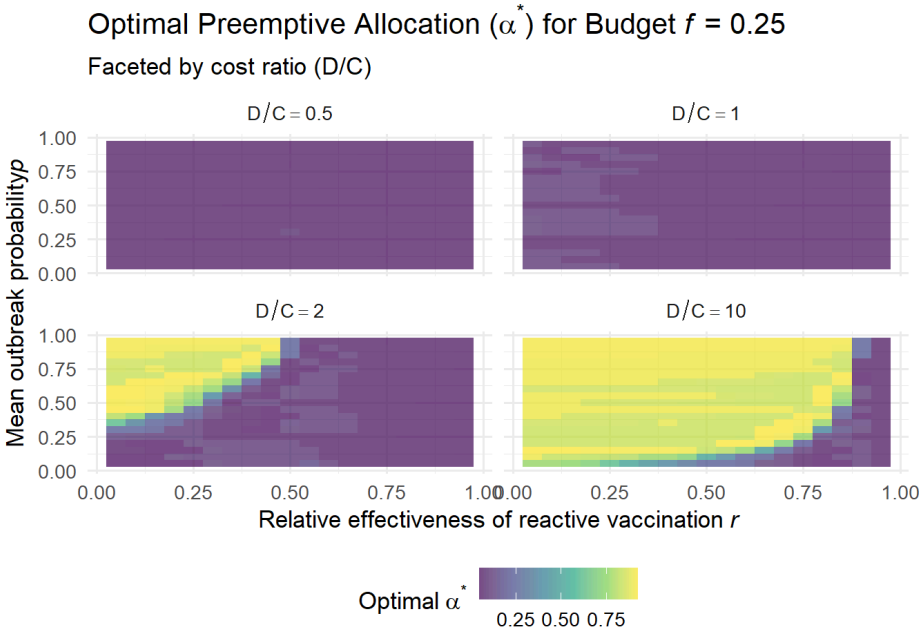
Parallel session although overhead is larger than the performance gain in this case

▶ Code

Optimal α for each parameter set

▶ Code

▶ Code



5.4 Imperfect assessment

5.4.1 Generating noisy ranks with a specified correlation to the original sequence

We aim to generate a noisy version of a standardized signal vector `p_z` such that the new vector `z_hat` has a target **Pearson correlation** r with `p_z`.

The following formula is used to construct the noisy vector:

$$\hat{z} = rp_z + \sqrt{1-r^2}\,\mathcal{N}(0,1)$$

where:

- p_z is a normalized version of the original signal (mean 0, variance 1)
- $\mathcal{N}(0,1)$ is an independent standard normal vector
- `r` is the desired correlation between $\hat{z}$ and p_z

This is a linear combination of two independent standard normal vectors:

$$Z_{\text{hat}} = r \cdot Z_{\text{signal}} + \sqrt{1-r^2} \cdot Z_{\text{noise}}$$

where:

- $Z_{\text{signal}} \sim \mathcal{N}(0,1)$
- $Z_{\text{noise}} \sim \mathcal{N}(0,1)$
- $\text{Cov}(Z_{\text{signal}}, Z_{\text{noise}}) = 0$

The variance of the resulting vector is:

$$\begin{aligned}\text{Var}(Z_{\text{hat}}) &= r^2 \cdot \text{Var}(Z_{\text{signal}}) + (1-r^2) \cdot \text{Var}(Z_{\text{noise}}) \\ &= r^2 + (1-r^2) = 1\end{aligned}$$

The covariance between `z_hat` and `p_z` is:

$$\text{Cov}(Z_{\text{hat}}, Z_{\text{signal}}) = r \cdot \text{Var}(Z_{\text{signal}}) = r$$

Therefore, the Pearson correlation is:

$$\text{Cor}(Z_{\text{hat}}, Z_{\text{signal}}) = \frac{\text{Cov}(Z_{\text{hat}}, Z_{\text{signal}})}{\sqrt{\text{Var}(Z_{\text{hat}}) \cdot \text{Var}(Z_{\text{signal}})}} = \frac{r}{1} = r$$

Once we know how to generate a sequence with a specified Pearson's r, given a sequence, we can also generate a sequence with desired Spearman's ρ because Spearman's ρ is a Pearson's r applied to ranked sequence. Similarly, using the relationship between Spearman's ρ, we can generate a sequence with a desired Kendall's tau.

5.4.1.1 test the function

▶ Code

```
[1] 0.7907704
```

▶ Code

```
[1] 0.5998717
```

References

Azman, Andrew S, Francisco J Luquero, Isabelle Ciglenecki, Rebecca F Grais, Amy Wesolowski, and Derek AT Cummings. 2019. "The Impact of Reactive Vaccination on Hepatitis e Outbreaks: A Modeling Study." *PLOS Neglected Tropical Diseases* 13 (10): e0007633. https://doi.org/10.1371/journal.pntd.0007633.

Colijn, Caroline, Colin Worby, and Marc Lipsitch. 2021. "What Is the Value of Information in Epidemic Modeling?" *Epidemics* 34: 100442. https://doi.org/10.1016/j.epidem.2021.100442.

Middleton, Fenner, and Daniel B Larremore. 2024. "Modeling the Transmission Mitigation Impact of Testing for Infectious Diseases." *Science Advances* 10 (eadd5108). https://doi.org/10.1126/sciadv.adk5108.

Quilty, Billy J, Sam Clifford, Joel Hellewell, et al. 2023. "Modeling Targeted Vaccination Strategies to Contain Mpox in Networks of High-Risk Groups." *Nature Communications* 14: 215. https://doi.org/10.1038/s41467-023-36177-5.

Wells, Chad R, Abhishek Pandey, Meagan C Fitzpatrick, Crystal Watson, et al. 2020. "Optimal Strategies for MERS-CoV Vaccination: A Modeling Study." *Lancet Infectious Diseases* 20 (10): e110–21. https://doi.org/10.1016/S1473-3099(20)30271-0.